

INNOVATION LAB BUILDOUT

DIORA

SAMA

JOSEPHINE

JACQUELINE

VIEW



INTERVIEW

MADDIE WISHART: UCSB MAKERSPACE MANAGER

- IYA is specialized / UCSB has a larger skill gap and smaller space → in parallel to innovation lab
- Trains student TAs
- RSVP system to manage chaos
- Tool suggestions:
 - 3D printing pens
 - Cricut
 - Button Maker

MOXIE CHILDREN'S MUSEUM



MAIN OBJECTIVES

MODULARITY | COLLABORATION | CONGESTION | INSPIRATION

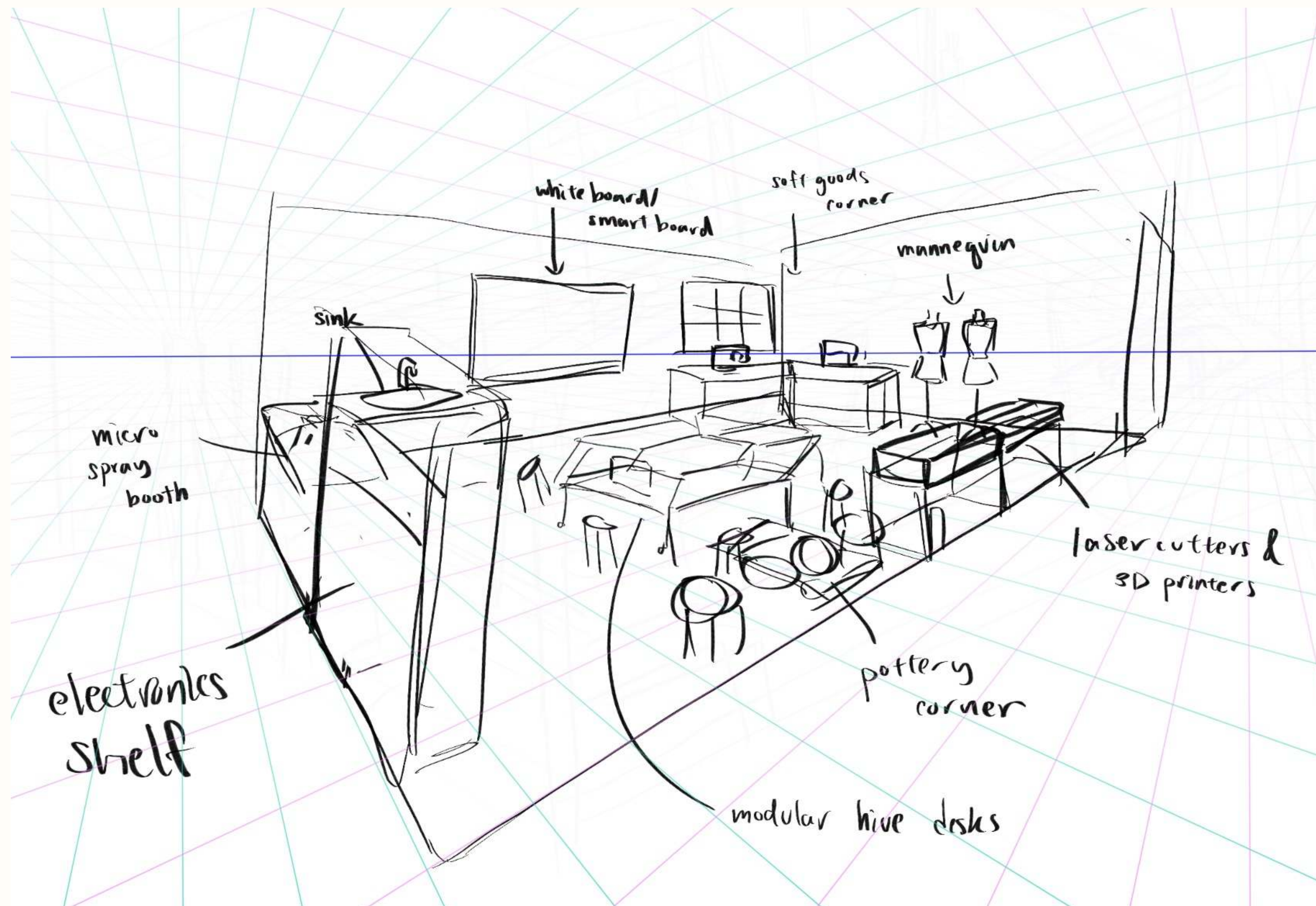
OPPORTUNITY STATEMENT

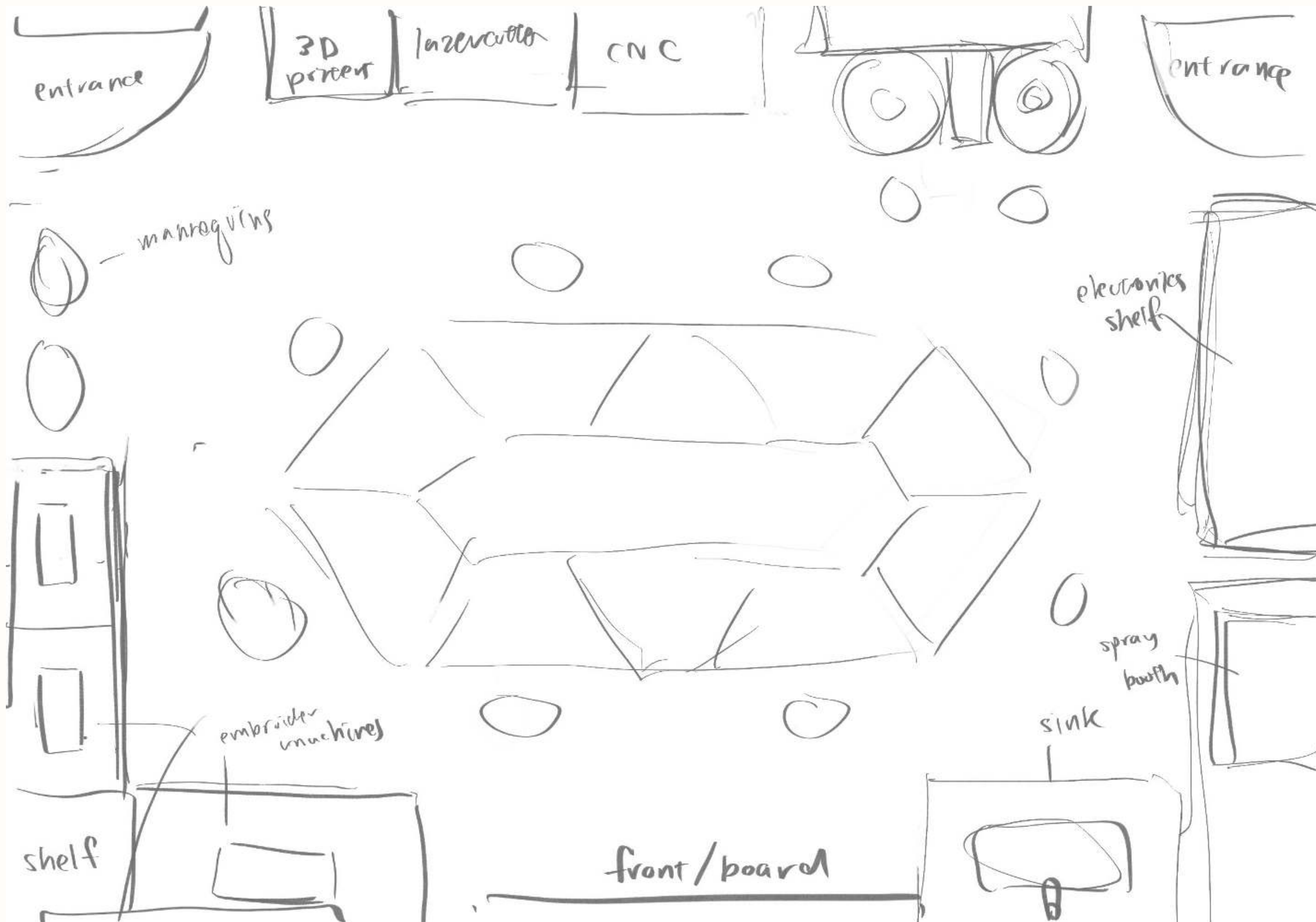
Currently, there is no **dedicated space** for students at SoLa tech center for the exploration of personal and collaborative **hardware and software projects**.

Our opportunity is in establishing a **makerspace** to address this gap, fostering **creativity, hands-on learning, and teamwork** amongst students."

MAKER SPACE

VISUALIZATIONS



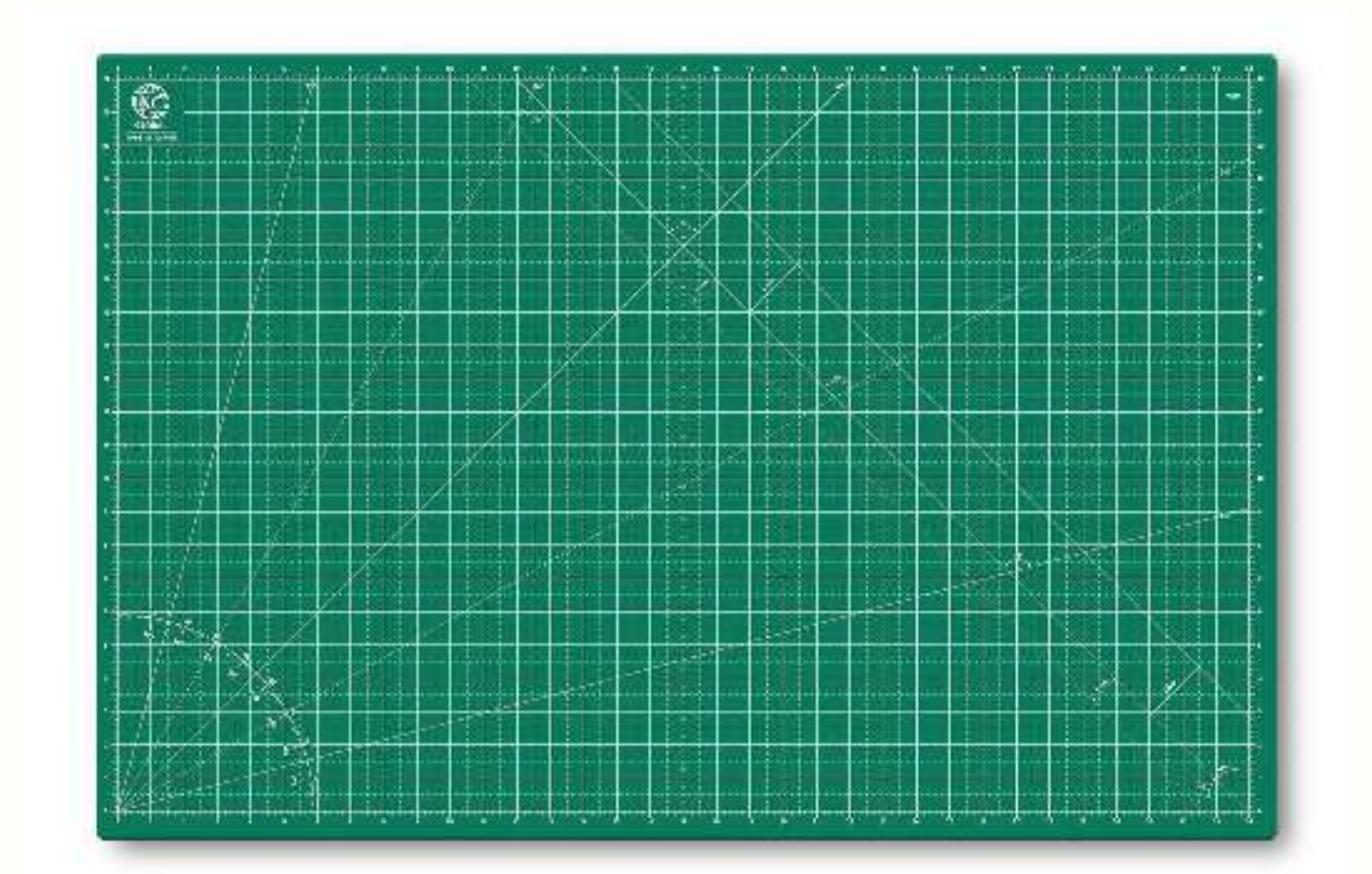


\$4.7K MODULAR DESKS



Modular Hive Tables (x6)

750 \$ Each



Cutting Mats (x6)

38 \$ Each

\$5K FURNITURE



Double Ed Makerspace Table with 6 Metal Stools 36"H
\$5,599.00



Rover Mobile Makerspace Powered Table with Open Double Storage
\$4,927.95



GangMei Metal Tool Storage Cabinet
\$269.00



IDEA Island Makerspace Table with 8 Stools by CEF 42"H
\$4,599.00



Horizon Makerspace Series Mobile Teachers Workstation with 9 Trays
\$2,123.55



Wall Control Pegboard Standard Tool Storage
\$116.48

\$1K

SOFT GOODS CORNER



SINGER 4452 Heavy Duty Sewing Machine

\$200 15.5" x 6.25" x 12"



Brother 1634DX 3/4 Thread Serger

\$250 11.73" x 10.98" x 13.19"



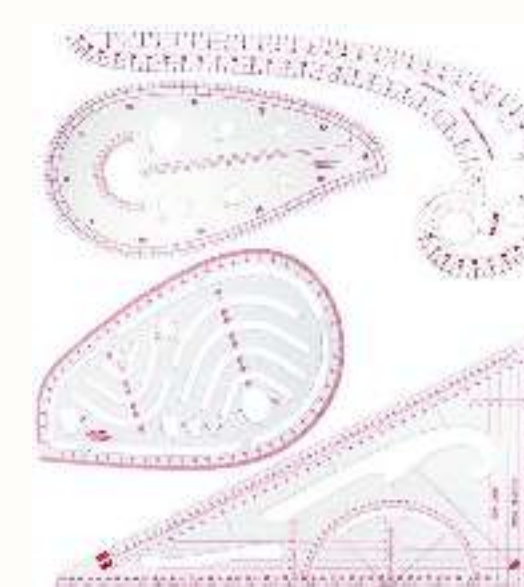
Brother SE700 Sewing and Embroidery Machine

\$500 15.5" x 6.25" x 12"



Dritz Adjustable Mannequin (Male & Female)

\$150-300 Adjustable sizing



Tools

- Rulers
- Tape (\$4) & plastic (\$4)
- Fabric Shears (\$15-\$30)
- Pencils
- Chalk (\$7)
- Curve tools (\$11)
- Pins & Clips (\$7)

\$1.6K

ELECTRONICS CORNER



Maxworks 80695 Bench Table Top 15-Bin Parts
\$24.99



YIHUA 1000B- 4 in 1 Station
\$399.99



ProTechTrader Make: Electronics 3rd Kit
\$281.99



FNIRSI-1014D Digital Oscilloscope
\$66.29



Electronic 72"W Workbench Tennsco Corp
\$1,199.99



Luckyrao DC Power Supply Adjustable Variable
\$66.99



Jameco BenchPro GTK-750A 95-Piece
\$224.95



Makey Makey Construction Kit
\$49.95

\$1.4K

PAIN AND POTTERY CORNER



Best Choice Products
Multipurpose Folding Table

\$130.99



Home Depot Pottery Wheel

\$36.99



Low Fire Pottery Clay

\$29.99



Air Dry Clay

\$29.99



Acrylic Paint

\$55.99

Sculpting Tools

\$7.64



Spray Paint (24)

\$189.99



Spray Paint Booth

\$866.00



Kiln

\$51.99

TOTAL INNOVATION LAB COSTS

MATERIALS \$22.1K

LABOR \$52/HR

SUPPORTING CURRICULUM



Bringing the curriculum to life

Making and fabrication.
The lab acts as a bridge
between theoretical learning
and practical applications.



Collaboration and knowledge sharing

Cross disciplinary collaboration
- facilitate interactions/ serves
as a melting pot where science,
technology, engineering, arts,
and mathematics blend



Testing and feedback

Students can test and refine
curriculum materials (like
simulations or activities) in
the lab, providing valuable
feedback to instructors.

SUPPORTING MATERIAL SCIENCE LIBRARY



Hands on Experiments/Prototyping

Allows students to investigate material properties and applications/limitations. This could include experiments on conductivity, strength, flexibility, and more.



Specialized workshops

The lab can host specialized workshops and seminars focused on specific topics within material science, such as nanotechnology, biomaterials, or polymers to provide students with more in-depth knowledge.

APPENDIX

STAFFING AND MAINTENANCE

1. Reservation/Checkout System for Tools and Materials

- **Purpose:**

Allows users to reserve and check out tools and materials for specific time slots. This system helps in managing the inventory and ensures that resources are efficiently utilized.

- **Features:**

- Online Booking: A user-friendly interface for reserving tools and materials. Should include images, descriptions, and availability.
- Calendar Integration: Displays availability and allows for booking directly through a calendar.
- Automated Notifications: Sends reminders to users about their reservations and due dates for returns.
- Reporting: Generates usage reports for better understanding and managing inventory.

- **Implementation:**

Can be developed in-house or using third-party booking systems customized for your needs. Consider platforms like Skedda, Cobot, or custom software with specific features for tool and material booking.

STAFFING AND MAINTENANCE

2. Database of Material + Tools

- **Purpose:**

To keep track of all materials and tools available in the maker space, including details like quantity, condition, and location.

- **Features:**

- Searchable Database: Easy-to-use search functionality for quickly finding tools and materials.
- Check-in/Check-out Tracking: Tracks when items are checked out and returned, including who has them.
- Maintenance Log: Records maintenance activities and schedules for each tool.
- Access Control: Different access levels for staff versus users.

- **Implementation:**

Use database management software that can be integrated with the reservation system. Consider using a web-based platform that can be easily updated and accessed by both staff and users.

STAFFING AND MAINTENANCE

3. Volunteer Student Assistants

- **Roles:**

Assisting in day-to-day operations, helping new users with tools, maintaining and organizing tools and materials, and ensuring rules are followed.

- **Recruitment and Training:**

- Recruitment: Through campus announcements, social media, and referrals. Look for students with a keen interest in making and helping others.
- Training: Must include safety training, customer service skills, and specific training on tools and materials management.

- **Scheduling and Management:**

Utilize a volunteer management system to schedule shifts, track hours, and communicate with volunteers.

STAFFING AND MAINTENANCE

4. Maker Space Manager

- **Responsibilities:**

- Overseeing the entire operation of the maker space.
- Staff and volunteer management.
- Budgeting and procurement of tools and materials.
- Ensuring safety protocols are followed.
- Community engagement and promoting the maker space.

- **Skills and Qualifications:**

Should have experience in managing similar spaces, technical knowledge of the tools and materials in the maker space, strong organizational skills, and the ability to work with a diverse group of individuals.

- **Hiring Process:**

Should include a thorough search to find a candidate with a blend of managerial skills and a passion for the maker movement. Consider utilizing professional networks, industry publications, and job boards that specialize in maker/creative industries.

STAFFING AND MAINTENANCE

5. Maintenance and Upkeep

- **Regular Maintenance Schedule:**

Establish a routine for checking and maintaining tools and equipment.

- **Repair and Replacement Plan:**

Have a budget and process in place for repairing or replacing tools and materials as needed.

- **Safety Inspections:**

Regular safety checks to ensure all equipment is in good working order and safe to use.